

CS787: GENERATIVE ARTIFICIAL INTELLIGENCE LANGUAGE MODELS

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Wed 12:00–13:15 at L20, Fri 12:00–13:15 at L19

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- Representation of *mental* form of *idea*

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 - **Statistical NLP**
- **Language model** predicts the *next word*
 - Coherence, grammaticality, fluency, (factuality)

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- For a sequence of words, *n-gram* probability is

$$P(w_1, w_2, \dots, w_l) = \prod_{i=1}^n P(w_i | w_{i-1}, w_{i-2}, \dots, w_{i-n+1})$$

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- **Interpolation** uses a weighted average of n -grams up to n

$$P = \lambda_1 P_{1\text{-gram}} + \lambda_2 P_{2\text{-gram}} + \cdots + \lambda_n P_{n\text{-gram}}$$

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- For a test sentence, a *better* model will predict better
- Correct word and sequence will have a *higher* probability
- **Perplexity** is the *inverse probability* of the test sequence
 - Per-token perplexity

$$\text{Perplexity}(W) = P(w_1 w_2 \dots w_n)^{-1/n} = \sqrt[n]{\prod_{i=1}^n P(w_i | \dots)^{-1}}$$

- Lower the perplexity, *better* the model

Example

- Unigram perplexity of string 0100
- If $P(0) = 0.5$, $P(1) = 0.5$, then perplexity is

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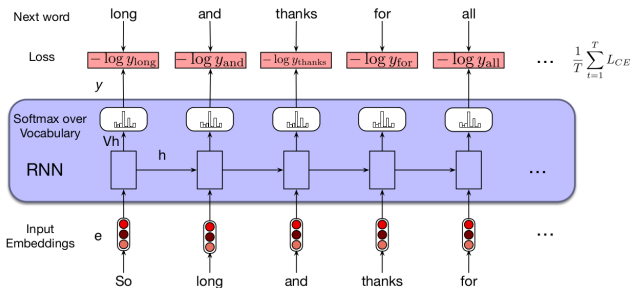
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- Second model is better
- Probabilities computed using training data
- Perplexity measured using test data
- Perplexity is connected to *cross entropy*

Language Modeling using RNN

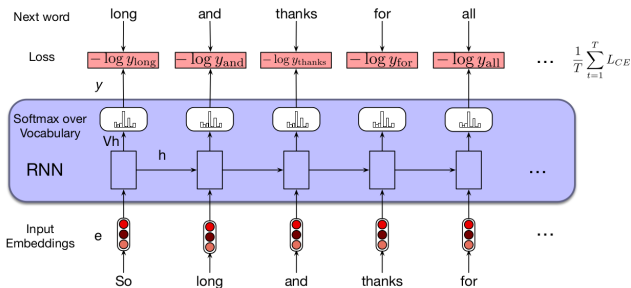


- Keep generating the next word for every time t
- Loss is average **cross-entropy loss**

$$\begin{aligned} L_{CE}(\hat{y}, y) &= - \sum_i y_i \ln \hat{y}_i \\ &= - \ln \hat{y}_i \text{ since } y_{i \neq k} = 0 \end{aligned}$$

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- Word embedding vectors can be one-hot or global (e.g., Word2Vec)